# Boston's Heat, Vulnerable Communities, and Open Space

A spatial analysis of Boston temperature distribution in relation to social vulnerability index factors and open space.

**by Lauren Tirona, Stephen Bond, Kimberly Wong, Sara Guefroudj, and Evan Lee**

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## *Question*

How are urban heat, affluence, and proximity to green spaces correlated in Boston, and furthermore, which areas and their inhabited demographics will be most affected by current projected heat due to climate change?

## *Background*

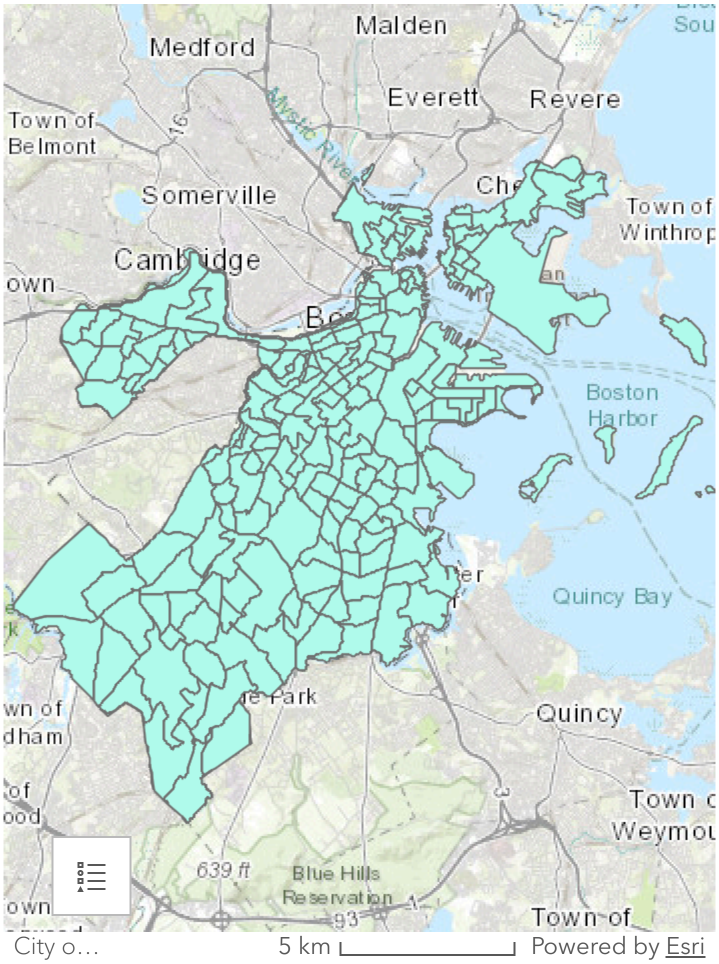
An urban heat island occurs when a city experiences much warmer temperatures than nearby rural areas. While the sun may hit an urban and rural area the same way, the surfaces in each environment absorb and hold heat differently. In rural areas, plants keep the air cool via transpiration. Urban areas, however, have few plants and are dominated by structures made of materials like steel and asphalt. These materials tend to be dark, and as a result, absorb more wavelengths of light and convert them into heat.

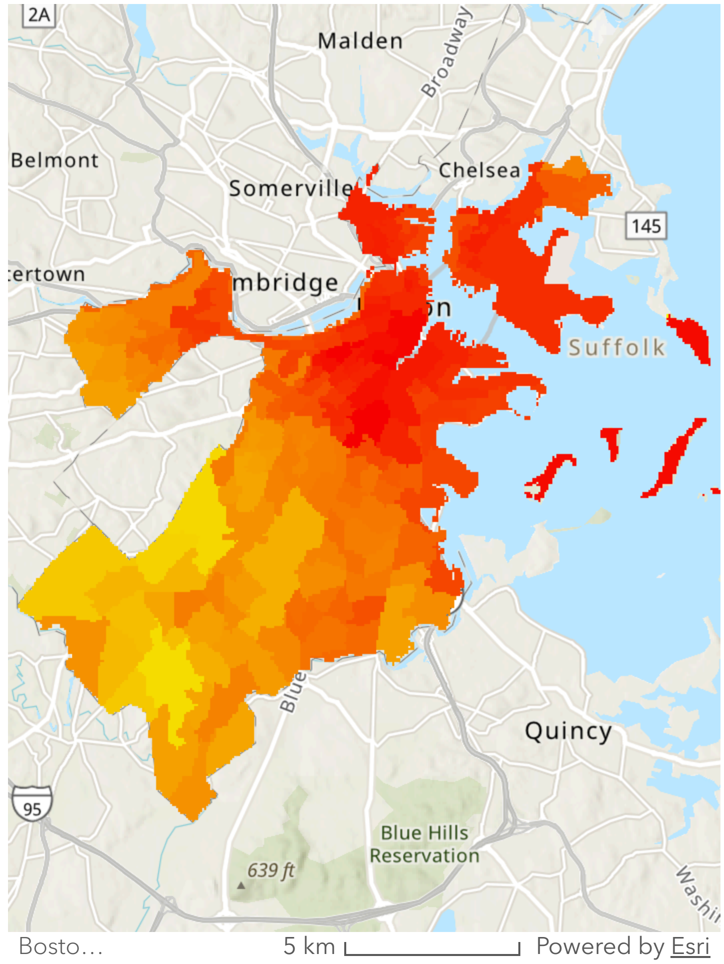
The area of focus for this project is Boston, Massachusetts. Since the early 1900s, Massachusetts has warmed by nearly 3.5°F, and models project a continued rise. Additionally, heat in Massachusetts comes with humidity, making it feel even hotter (Massachusetts Department of Public Health, 2026). This issue underscores the importance of green spaces in urban areas.

Rising temperatures in urban areas is a climate issue, but just as much a social justice issue. One study that utilized remote sensing in Boston found a significant relationship between Home Owners' Loan Corporation designation and mean summer temperature, and further, that Black and Hispanic people are overrepresented in areas with uniquely high temperatures (Gray, 2024). Another study examining the distributional equity of urban vegetation in 10 US urbanized areas found a strong association between income and higher education and urban vegetation (Nesbitt et al., 2018). Increasing temperatures, exacerbated in urban areas with little to no green spaces, exposes residents to extreme heat and its consequential health effects.

## *Layers*

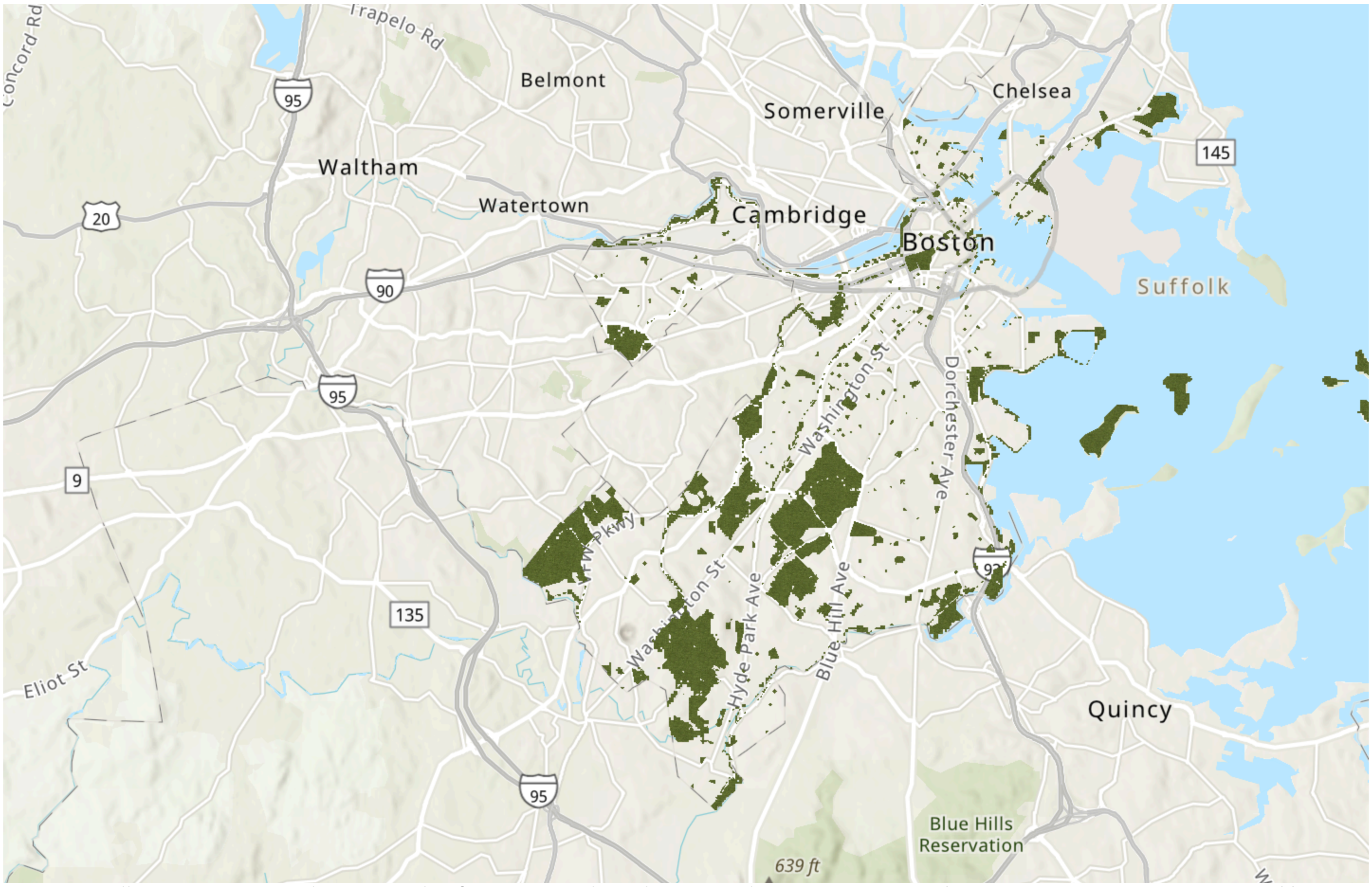
To explore how urban heat, social vulnerability indicators and proximity to open spaces are correlated, the following social vulnerability factors were analyzed across Boston. This includes medical illness, age, poverty level, and race.




## Temperature

This map shows the distribution of heat across Boston. Each polygon represents the mean morning temperature in Fahrenheit within the census tract boundaries. Utilizing the Polygon to Raster geoprocessing tool, we were able to make a continuous visual model of heat throughout the city. We additionally used the Project Raster tool to reproject this layer using the same coordinate system as the rest of our data (WGS-1984).

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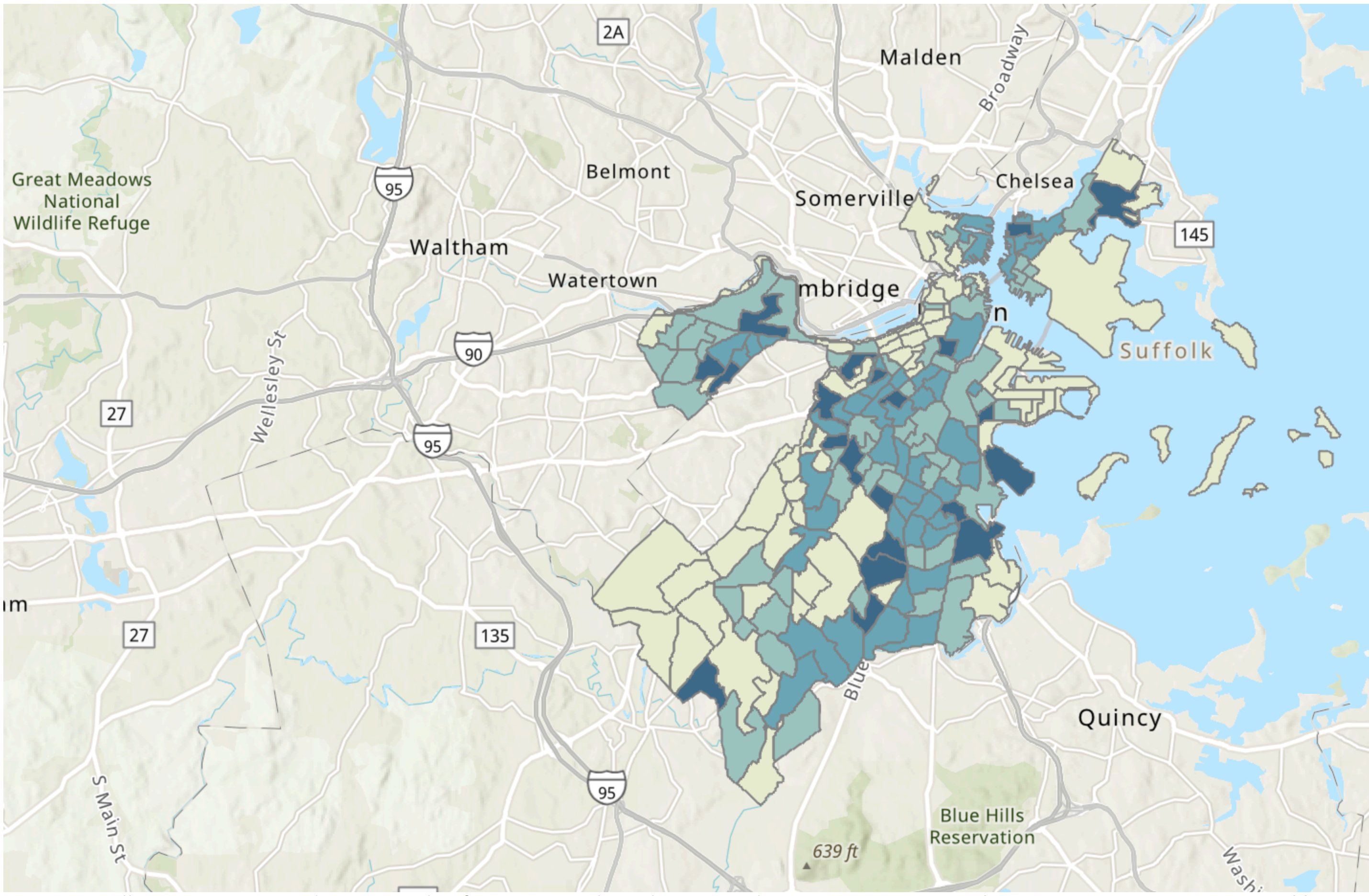
## Open Space

This map shows the distribution of open spaces across Boston. Each polygon measures open space area in acres.

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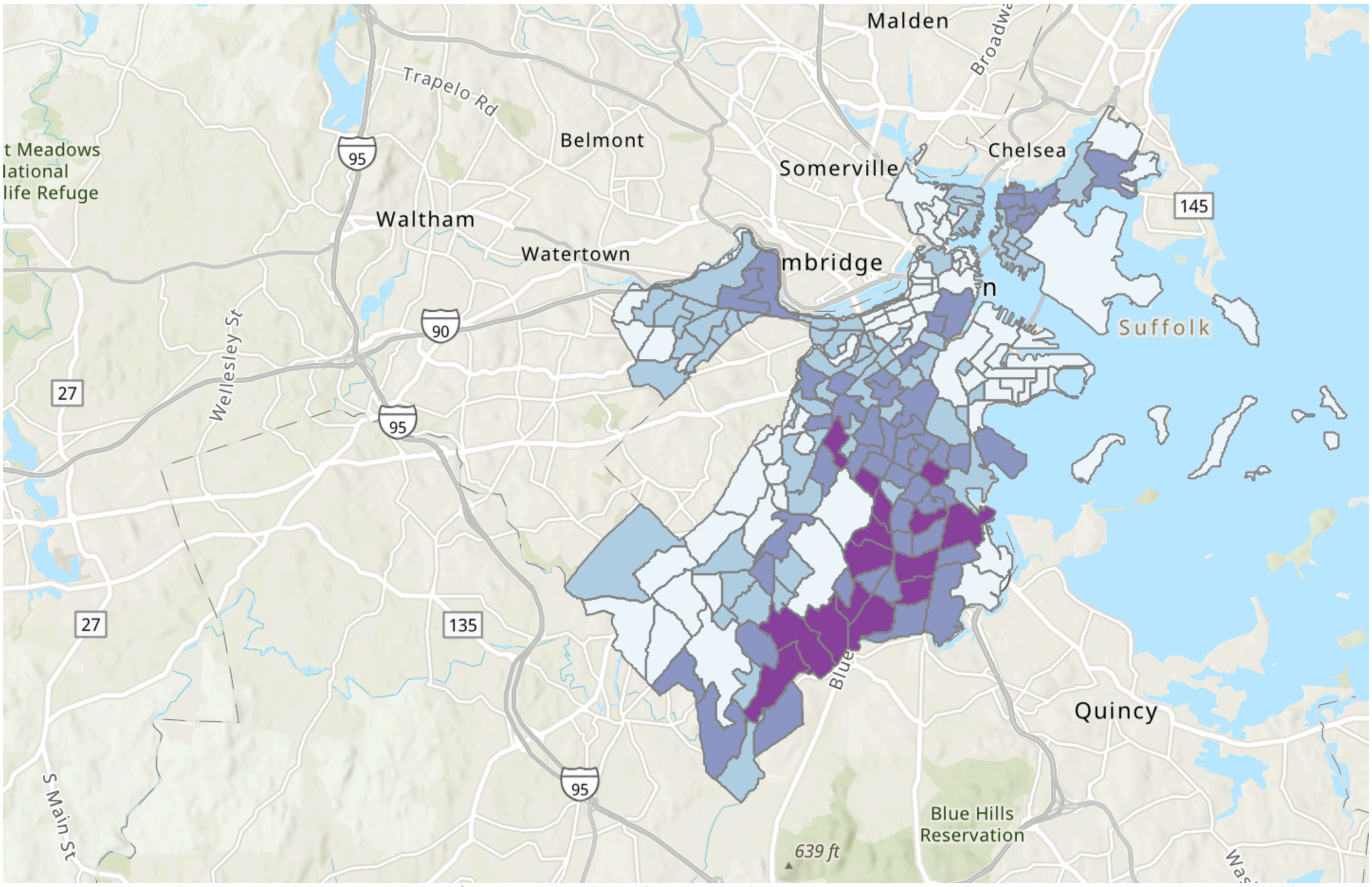
## Low to no Income

This map shows the distribution of income across Boston. Each polygon represents the number of people living at or below 149% of poverty level within each census tract(2010)

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### People of Color

This map deals with the number of individuals that are Black, Native American, Asian, Island, Other, Multi, and Non-white Hispanics within each census tract(2010)

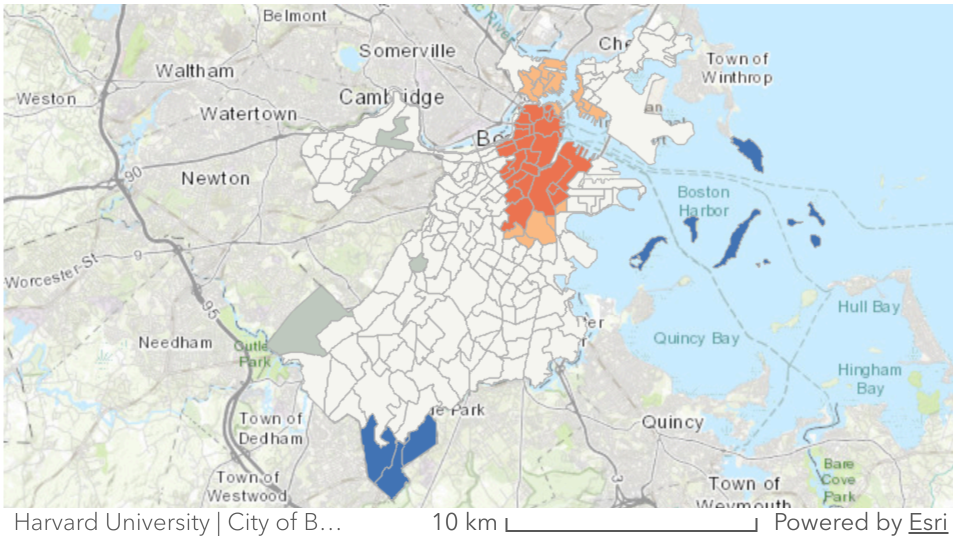
**Older Adult**

This map highlights the number of adults over the age of 65 within each census tract(2010)

**Medical Illness**

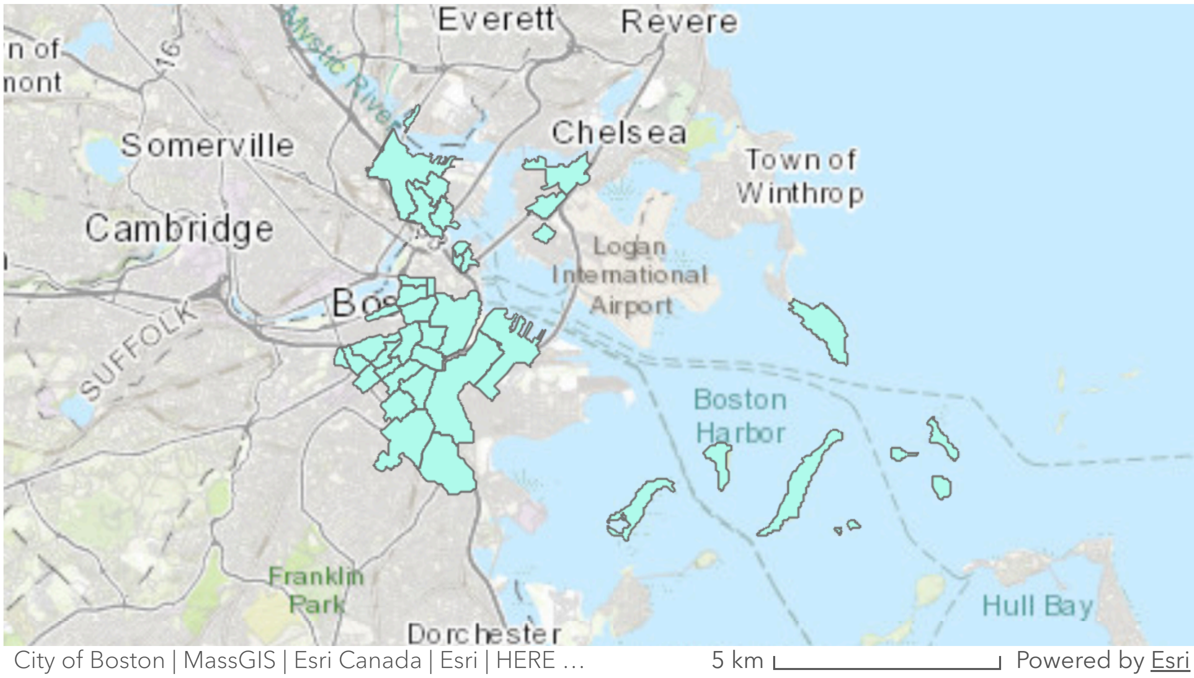
This map highlights the estimated count of residents with a medical illness(sum of asthma in children, asthma in adults, heart disease, emphysema, bronchitis, cancer, diabetes, kidney disease, and liver disease) within each census tract(2010)

## *Visualizing Extreme Temperature, Open Space, and Vulnerability*




Temperature Hot Spot Analysis

We performed a Hot Spot Analysis to identify statistically significant hot spots and cold spots. We observed that hot spots with high confidence appear to generally occur in the North End, the South End, and Roxbury. This analysis acted as a precursive step that allowed us to confirm that certain parts of Boston have significantly high temperatures as opposed to a random distribution of high temperatures throughout the city. Following this finding, we decided to investigate the overlap of these hotter areas with areas of extreme social vulnerability through spatial join analyses.




Hottest Areas of Boston ≥ 76.5°F

To determine the hottest districts of Boston, we used the "Select Layer by Attribute" tool to select and create a layer of the districts that had an average temperature greater than or equal to 76.5°F and found 40/211 areas that fit the description.

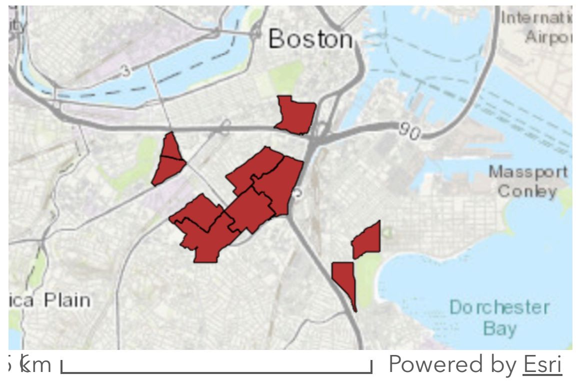
We also used the tool to identify the districts in Boston with the largest shares of each social parameter, creating four different layers for each factor. The first layer depicted areas that had a number of people of color greater than or equal to 2,700, which the "Select Layer by Attribute" tool found to be 47 areas. The second layer displayed areas that had greater than or equal to 1,500 low- to no-income individuals, which the tool determined to be 46 areas. We then created a third layer to show 14 areas that had greater than or equal to 750 older adults. Finally, the last layer pictured 28 areas that had a number of medically ill individuals greater than or equal to 1,900.

Next, we used the "Spatial Join" tool to determine where each of the social vulnerability factors overlaps with heat. For the parameters, we set the respective social vulnerability layer as the target feature and the new temperature layer depicting the hottest areas of Boston as the join feature, used a one-to-one join operation, and an intersect match option to separately


Spatial Join of High Temperature and High POC

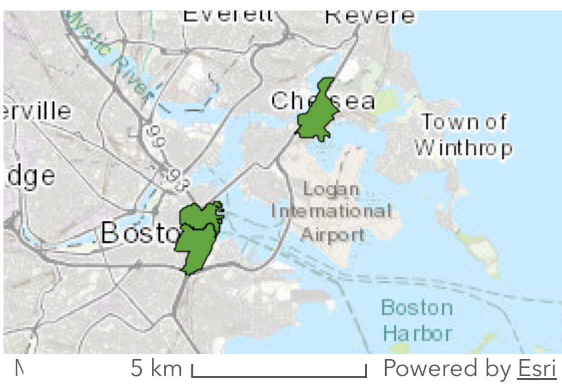
determine which areas had both increased heat and a high number of each of social vulnerability factors.

We found 8 districts with increased heat and people of color and 14 districts with increased heat and low- to no-income individuals.




Spatial Join of High Temperature and High Poverty

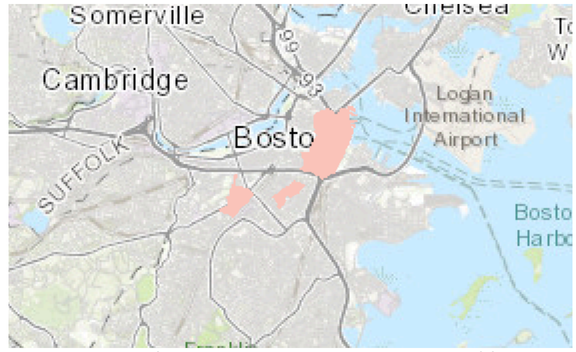
By spatially joining layers of Boston's heat map with Social Vulnerability Index (SVI) parameters, we identified critical intersections where extreme heat overlaps with at-risk populations.  High-heat areas showed a strong overlap with areas reporting high concentrations of poverty and people of color (POC), most notably in Roxbury and East Boston. From an environmental justice standpoint, it is essential that policymakers and officials invest resources into these communities because not only will they be even hotter, but especially because the high concentration of low-income residents may lack the resources to mitigate the heatwaves.



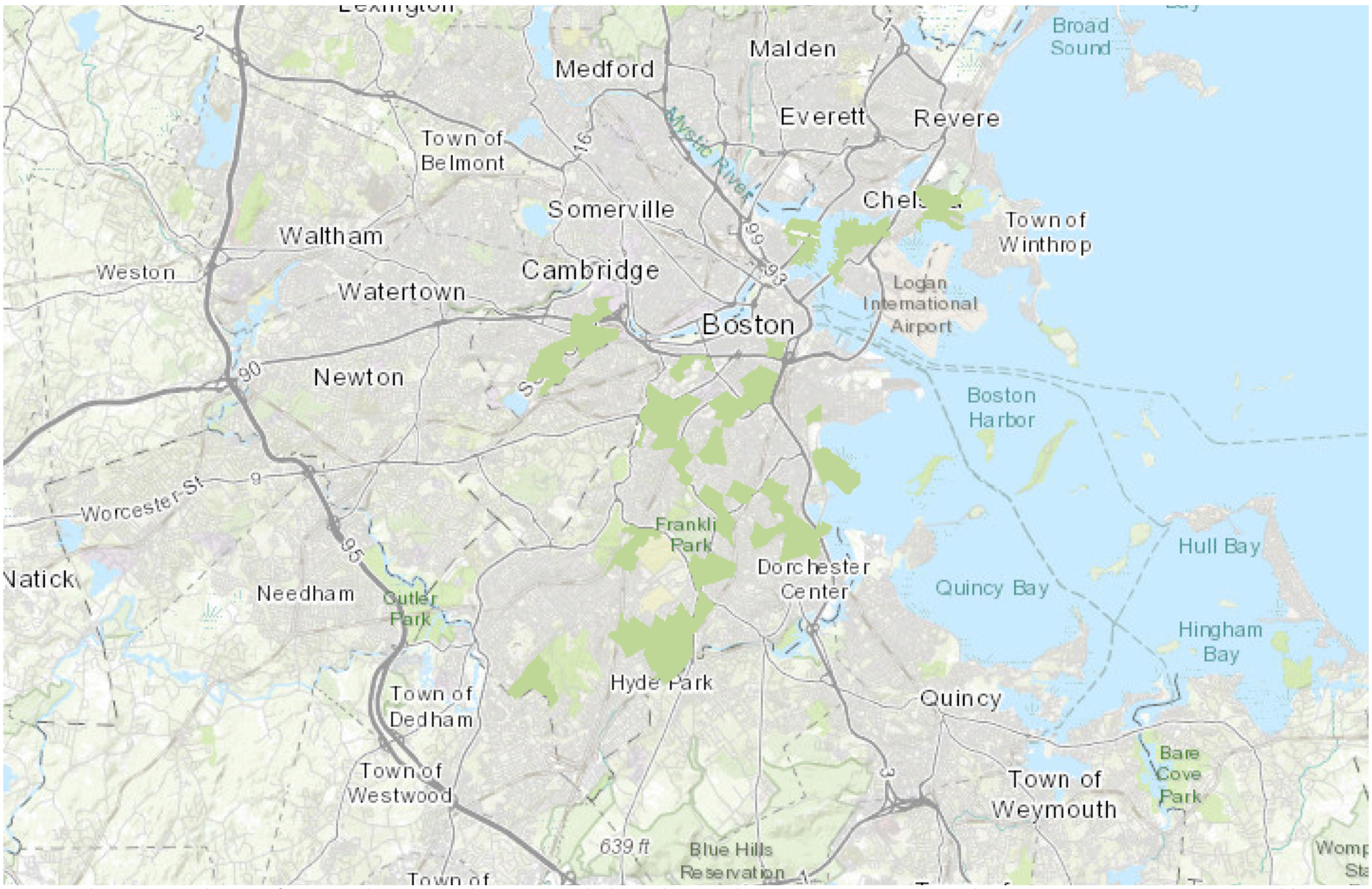

Spatial Join of High Temperature and High Older Adults

After running these spatial joins, we determined 3 districts with both increased temperatures and a high number of older adults, as well as 6 districts with increased temperatures and medically ill individuals.

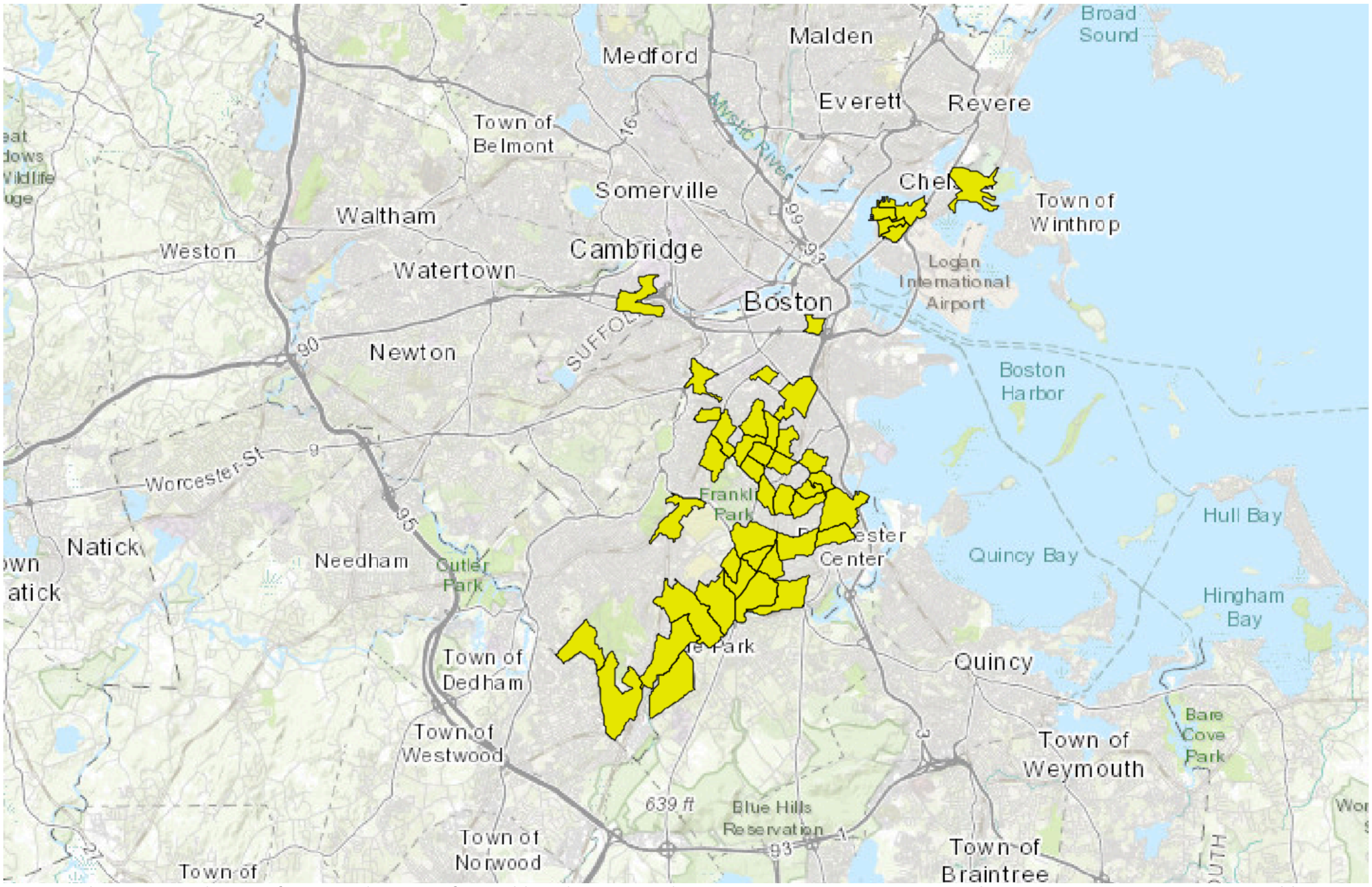
A similar trend was observed regarding age and medical illnesses. Our analysis showed a significant spatial correlation between high-temperature zones and residents with pre-existing medical illness and old age. Specifically, the North End and the Leather District were found to be primary hotspots for elderly and medically vulnerable populations. This spatial trend goes southward through South Boston and the Northeastern University area, continuing into Downtown and the Boston Common as well. Just as mentioned before, areas like North End and the Leather district should receive extra attention for heatwave response because of not only the high temperatures, but also the high concentration of vulnerable Bostonians.




Spatial Join of High Temperature and High Medically Ill Individuals




# Spatial Join of Open Spaces and High Poverty

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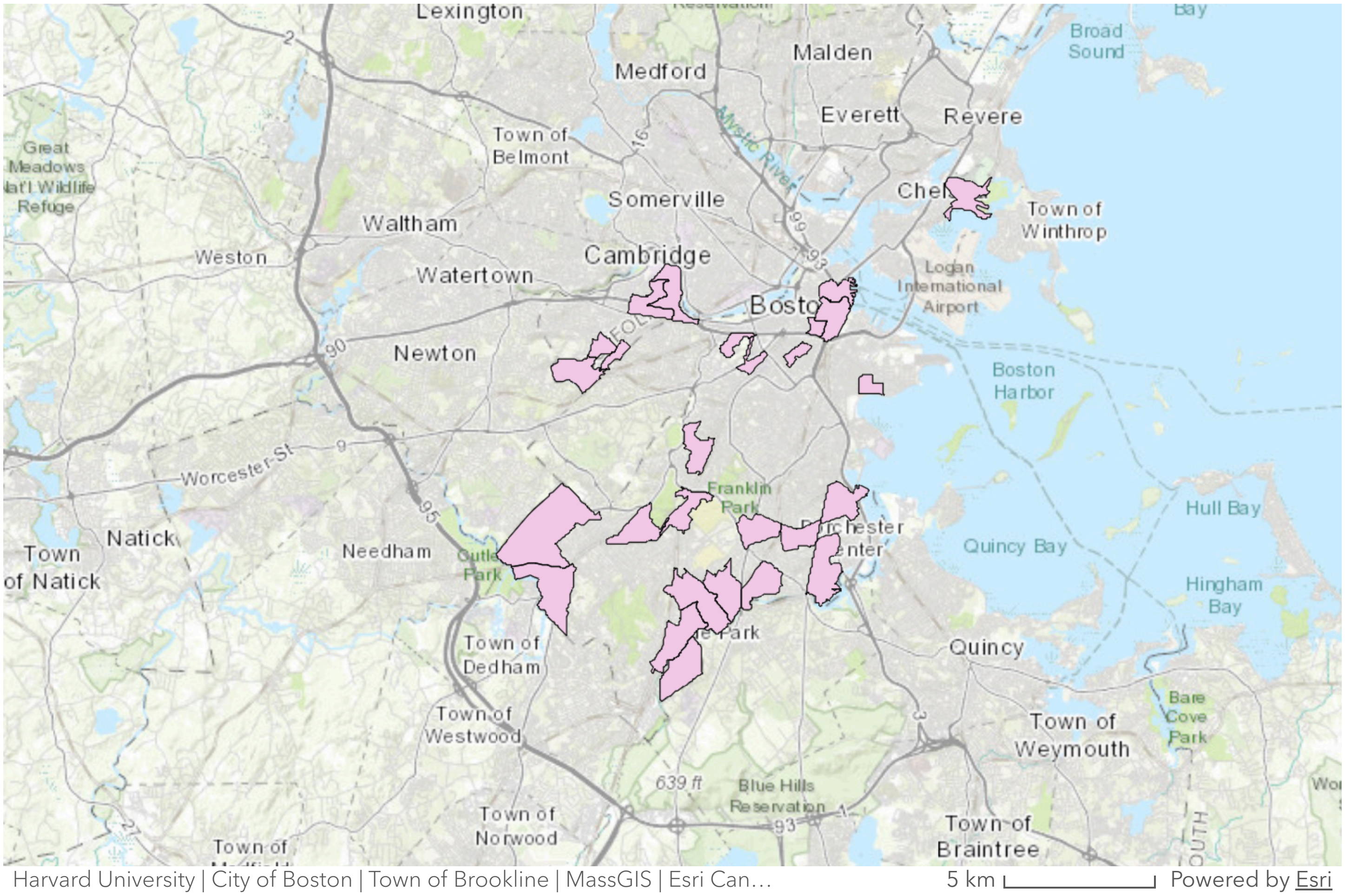
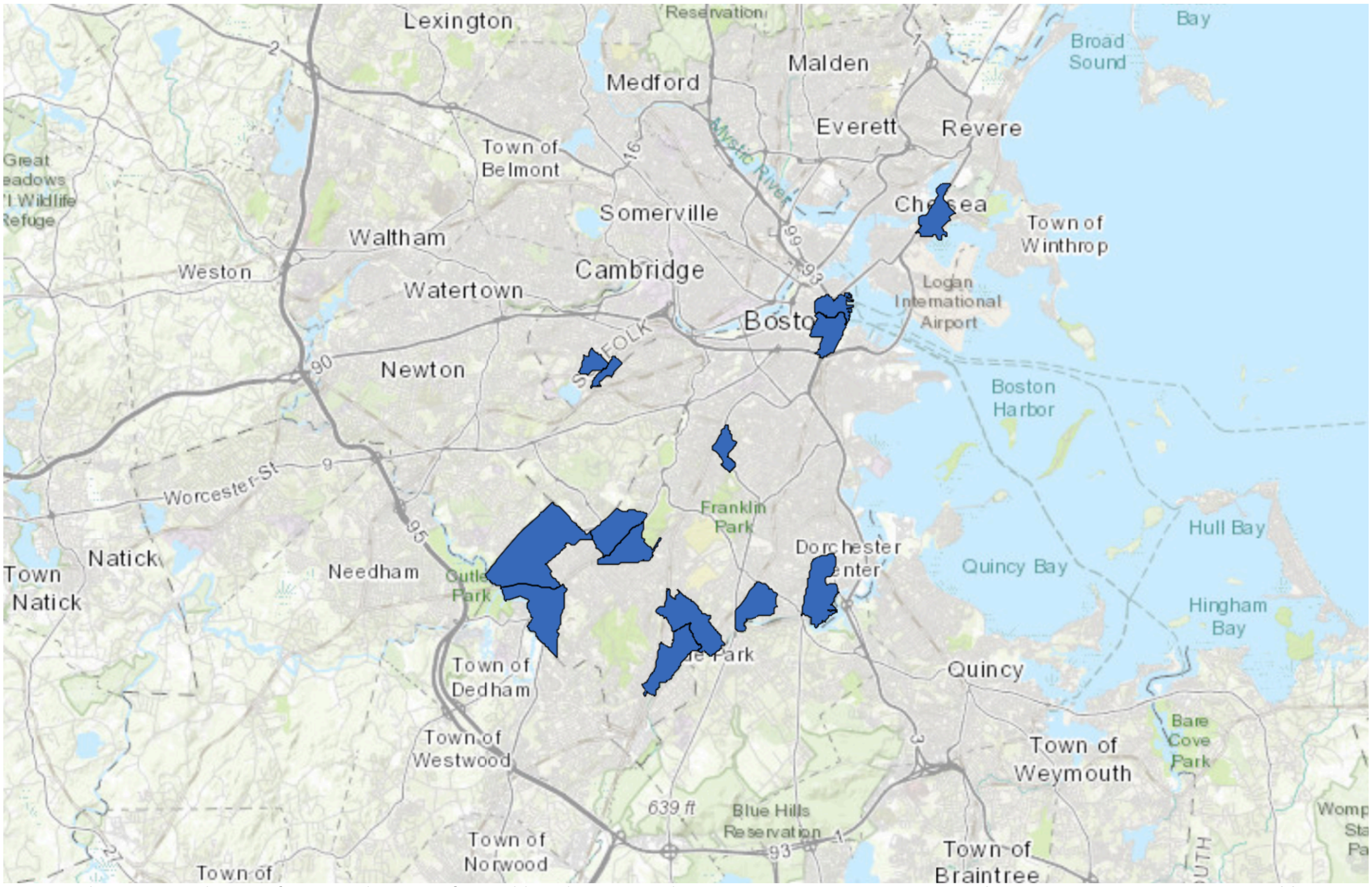
## Spatial Join of Open Spaces and High POC




## Spatial Join of Open Spaces and High Medically Ill Individuals




## Spatial Join of Open Spaces and High Older Adults

We also used the "Spatial Join" tool to determine the areas with overlap in open spaces and the social vulnerability parameters. We ran these using the original "Open Spaces" layer, new social vulnerability layers depicting the areas with increased numbers of each factor, a one-to-one join operation, and intersect match option. We found 42 districts with overlap in open spaces and increased low- to no-income individuals, 43 districts with overlap in open spaces and increased people of color, 27 districts with overlap in open spaces and increased medically ill individuals, and 14 districts with overlap in open spaces and increased older adults.

When it comes to open space, we could not find any hotspots or clear patterns as to where large amounts of ill people resided next to open spaces besides in the Allston-Brighton area, however the good news is that there are many of these Boston areas. When it comes to elderly residents being proximal to open spaces, there were a lot in West Roxbury. The Roxbury area, in general, is also where lots of people of color and low-income families are next to urban open spaces, which is good news from an environmental justice perspective. Other areas that were on these social vulnerability indexes with close proximity to open spaces included East Boston and Dorchester, as well as other neighborhoods throughout the city.

*Statistical Analysis*

## Finding Mean Open Space and Mean Temperature

While a polygon layer depicting temperature distribution and open space across Boston was found, the polygon shapes and projections did not align with those provided by the social vulnerability index. To address this discrepancy, the Zonal Statistics as Table geoprocessing tool was run, calculating average temperature values and open-space measurements for each neighborhood polygon in the social vulnerability index. These mean-value polygons will be essential when conducting more in-depth spatial analysis, as they help minimize the likelihood that these processes introduce distortion and violate the assumption that all variables are measured on the same spatial scale, leading to inaccurate findings.

## Running a Generalized Linear Regression

Two Generalized Linear Regression (GLR) models were fitted to our data to identify significant relationships between the variables temperature, open space, and the social vulnerability indices. The first linear regression used mean temperature as the dependent variable and open space, old age, medical illness, people of color, and low-to-no income as explanatory variables. Since all polygons were assigned to the same-sized polygons and projections due to previous geoprocessing, the regressions ran successfully. Interestingly, none of the explanatory variables had a statistically significant influence on the dependent variable, as all of the p-values exceeded 0.05. However, the coefficients still provide valuable insight. When examining intercept values for older adults and people of color, and the mean open space area, these variables are negatively associated with the mean temperature. Whereas, the intercepts for medical illness and low-to-no income show a positive relationship with the mean temperature.

**Summary of GLR Results [Model Type: Count (Poisson)]**

| Variable | Coefficient[a] | StdError | z-Statistic | Probability[b] | VIF[c] |
|---|---|---|---|---|---|
| Intercept | 4.329966 | 0.022302 | 194.148159 | 0.000000* | -------- |
| OLDERADULT | -0.000019 | 0.000049 | -0.397348 | 0.691111 | 1.887677 |
| LOW_TO_NO | 0.000004 | 0.000019 | 0.189342 | 0.849825 | 2.382722 |
| POC2 | -0.000003 | 0.000009 | -0.325549 | 0.744766 | 2.290372 |
| MEDILLNES | 0.000003 | 0.000022 | 0.126498 | 0.899338 | 2.475239 |
| OS_MEAN | -0.000120 | 0.000187 | -0.638844 | 0.522924 | 1.089952 |

**GLR Diagnostics**

| | | | |
|---|---|---|---|
| Input Features | c7230a7a-4081-4743-b911-e18f66e1beca2020330-1-17gw6be.a4ds | Dependent Variable | HEAT_MEAN |
| Number of Observations | 180 | Akaike's Information Criterion (AICc)[d] | 1123.000000 |
| Average Count | 75.327778 | Deviance Explained[e] | 0.245923 |
| Joint Wald Statistic[f] | 0.776493 | Prob(>chi-squared), (5) degrees of freedom | 0.978508 |

GLR Results when Dependent Variable was Mean Temperature

The second linear regression utilized mean open space area as the dependent variable and mean temperature, old age, medical illness, people of color, and low to no income as the explanatory variables. All the explanatory variables were found to have a statistically significant influence on the mean open space value, demonstrated by all p-values being less than 0.05. When observing the coefficients, we can see that old age, low-to-no income, and mean temperature all have negative relationships to mean open space. The mean temperature has the greatest influence on mean open space, as the mean temperature increases by a factor of 1, the open space decreases by 0.61 acres. The variables, people of color and medical illness, have small but positive relationships with open space. Attached below are the GLR Results and the residual map.

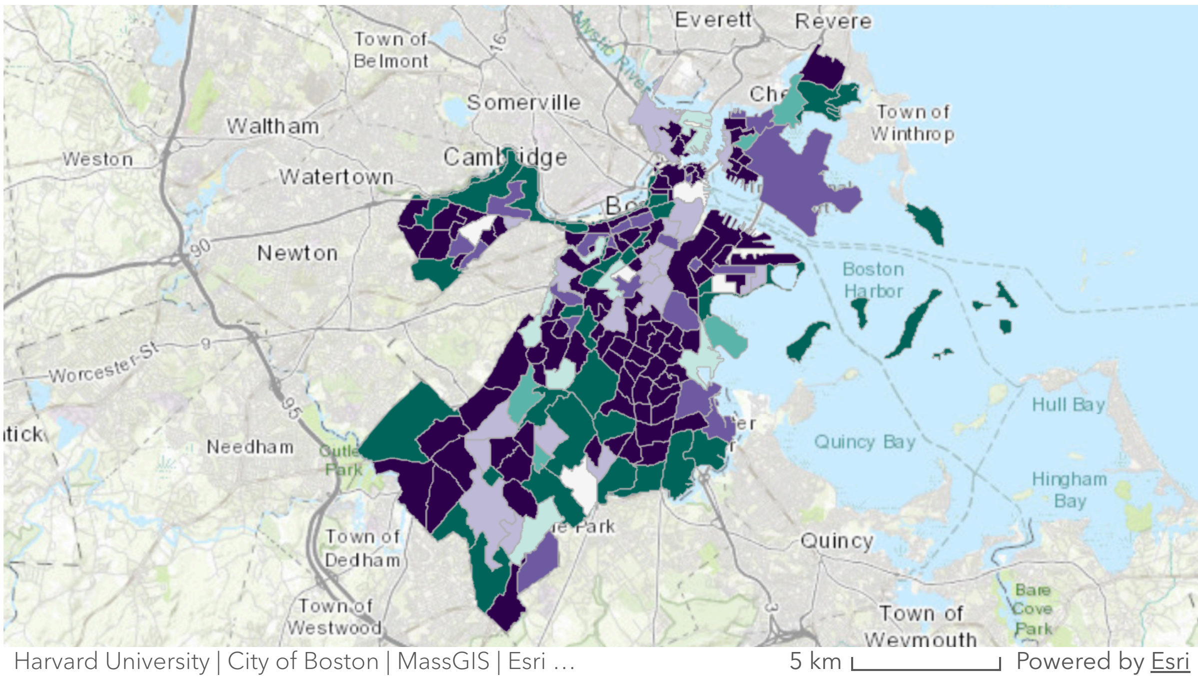
**Summary of GLR Results [Model Type: Count (Poisson)]**

| Variable | Coefficient[a] | StdError | z-Statistic | Probability[b] | VIF[c] |
|---|---|---|---|---|---|
| Intercept | 49.572154 | 1.103946 | 44.904511 | 0.000000* | -------- |
| OLDERADULT | -0.001864 | 0.000091 | -20.543148 | 0.000000* | 1.948797 |
| LOW_TO_NO | -0.000957 | 0.000044 | -21.841732 | 0.000000* | 2.366934 |
| POC2 | 0.000170 | 0.000019 | 9.063019 | 0.000000* | 2.415598 |
| MEDILLNES | 0.000375 | 0.000037 | 10.055171 | 0.000000* | 2.487783 |
| HEAT_MEAN | -0.610127 | 0.014876 | -41.014822 | 0.000000* | 1.134863 |

**GLR Diagnostics**

| | | | |
|---|---|---|---|
| Input Features | c7230a7a-4081-4743-b911-e18f66e1beca2020330-1-17gw6be.a4ds | Dependent Variable | OS_MEAN |
| Number of Observations | 180 | Akaike's Information Criterion (AICc)[d] | 7330.000000 |
| Average Count | 25.561111 | Deviance Explained[e] | 0.345965 |
| Joint Wald Statistic[f] | 3552.499513 | Prob(>chi-squared), (5) degrees of freedom | 0.000000* |

GLR Results when Dependent Variable was Mean Open Space




Open Space GLR Deviance Residuals

## Spatial Autocorrelation

While the mean temperature Generalized Linear Regression did not result in sufficient residuals of over- and under-predictions, the mean open space Generalized Linear Regression produced a complex deviance residual map. Using this residual map, a spatial autocorrelation (Global Moran's I) was run to investigate whether these errors were spatially clustered, dispersed, or random. The Moran's Index of -0.005 indicates these residuals have no autocorrelation and the spatial distribution is random. If the value were significantly greater than 0, it would indicate clustering, and if the value were significantly less than 0, it would indicate a greater dispersion. The p-value of 0.978 is greater than 0.005, meaning the results are not statistically significant, and we can not reject the null hypothesis that the data is randomly distributed. Finally, when taking into consideration the z-score of 0.0274, the idea that these residuals are randomly distributed is further supported. These values demonstrate that the model does not omit any influence from different spatial variables and adequately accounts for spatial data.

**Global Moran's I Summary**

| | |
|---|---|
| Moran's Index | -0.005146 |
| Expected Index | -0.005587 |
| Variance | 0.000258 |
| z-score | 0.027440 |
| p-value | 0.978109 |

Global Moran's I Results

## *Conclusion*

We ultimately concluded that there is a statistically significant relationship between our chosen social vulnerability factors– people of color, low to no income, older adults, and medical illnesses– and open space. An important finding in regards to environmental justice is that old age and low to no income are both negatively correlated with open space. While we did not determine statistical significance for the relationships between temperature and social vulnerability factors, we did see a negative correlation between temperature and open space (p=0.000000).

A limitation of this study is its assessment of the future impacts of climate change, as high-resolution models predicting temperature increases are not widely available to the public. In the future, we would like to incorporate these projected temperatures to understand the potential impacts of climate change on different neighborhoods in Boston and evaluate whether the relationships between social vulnerability variables, heat, and open spaces change over time. We would also like to consider other social vulnerabilities, including but not limited to the level of education and disabilities.

Furthermore, while this storymap effectively demonstrates the relationship between areas experiencing higher temperatures, open space, and larger disadvantaged communities, these findings can be utilized for urban planning strategies. The results, for example, can help identify locations where the implementation of cooling factors like more open space, bodies of water, or cooling stations would be most impactful. Lastly, we would like to acknowledge that there may be microspatial temperature variations, as well as more profound temperature variations in the greater Massachusetts region, which are not captured by our model but are scales that this project could replicate, given adequate data.

# *Sources*

Gray, L. (2024). Remote Sensing-Based Analysis of Urban Heat Islands and Historical Housing Discrimination In Boston, MA. Dartmouth Digital Commons. https://digitalcommons.dartmouth.edu/geography_senior_theses/8/

*Massachusetts Unhealthy Heat Forecast*. (2025). Mass.gov. https://www.mass.gov/info-details/massachusetts-unhealthy-heat-forecast

Chen, J. T., Rehkopf, D. H., Waterman, P. D., Subramanian, S. V., Coull, B. A., Cohen, B., Ostrem, M., & Krieger, N. (2006). Mapping and Measuring Social Disparities in Premature Mortality: The Impact of Census Tract Poverty within and across Boston Neighborhoods, 1999–2001. Journal of Urban Health, 83(6), 1063–1084. https://doi.org/10.1007/s11524-006-9089-7

**Open Space Layer** *Open Space - Analyze Boston*. (2024). Boston.gov. https://data.boston.gov/dataset/open-space

| | |
|---|---|
| **Social Vulnerability Layer** | *Climate Ready Boston Social Vulnerability - Analyze Boston*. (2017). Data.boston.gov. https://data.boston.gov/dataset/climate-ready-boston-social-vulnerability |
| **Temperature Layer** | *CT Mean Heat Index - Analyze Boston*. Boston.gov. https://data.boston.gov/dataset/ct-mean-heat-index |